

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	4	9	18

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

A-1 We need to minimize objective function

$$Q = \sum \epsilon_i^2$$

$$\text{So, } Q = (y_1 - \theta_1 - \theta_2)^2 + (y_2 - \theta_1 + 2\theta_2)^2 + (y_3 - 2\theta_1 + \theta_2)^2$$

Now ~~for~~ LSE ~~of~~ θ_1 , we need $\frac{\partial Q}{\partial \theta_1} = 0$

$$\Rightarrow -2(y_1 - \theta_1 - \theta_2) - 2(y_2 - \theta_1 + 2\theta_2) - 4(y_3 - 2\theta_1 + \theta_2) = 0$$

~~$$-2y_1 + 2\theta_1 + 2\theta_2 - 2y_2 + 2\theta_1 - 4\theta_2 - 4y_3 + 8\theta_1 - 4\theta_2 = 0$$~~

$$\Rightarrow -2y_1 + 2\theta_1 + 2\theta_2 - 2y_2 + 2\theta_1 - 4\theta_2 - 4y_3 + 8\theta_1 - 4\theta_2 = 0$$

$$\Rightarrow 12\theta_1 - 6\theta_2 = 2y_1 + 2y_2 + 4y_3 \quad \text{--- (1)}$$

For LSE of θ_2 , we need $\frac{\partial Q}{\partial \theta_2} = 0$

$$-2(y_1 - \theta_1 - \theta_2) + 4(y_2 - \theta_1 + 2\theta_2) + 2(y_3 - 2\theta_1 + \theta_2) = 0$$

$$\Rightarrow -2y_1 + 2\theta_1 + 2\theta_2 + 4y_2 - 4\theta_1 + 8\theta_2 + 2y_3 - 4\theta_1 + 2\theta_2 = 0$$

$$\Rightarrow -6\theta_1 + 12\theta_2 = 2y_1 - 4y_2 - 2y_3 \quad \text{--- (2)}$$

$$\text{So, } 12\hat{\theta}_1 - 6\hat{\theta}_2 = 2y_1 + 2y_2 + 4y_3$$

$$-6\hat{\theta}_1 + 12\hat{\theta}_2 = 2y_1 - 4y_2 - 2y_3$$

$$\textcircled{1} + 2 \times \textcircled{2}$$

$$18\hat{\theta}_2 = 6y_1 - 6y_2$$

$$\Rightarrow \hat{\theta}_2 = \frac{6y_1 - 6y_2}{18} = \frac{y_1 - y_2}{3}$$

$$12\hat{\theta}_1 - 2(y_1 - y_2) = 2y_1 + 2y_2 + 4y_3$$

$$12\hat{\theta}_1 = 2y_1 - 2y_2 + 2y_1 + 2y_2 + 4y_3$$

$$= 4(y_1 + y_3)$$

$$\boxed{\hat{\theta}_1 = \frac{y_1 + y_3}{3}}$$

$$\& \boxed{\hat{\theta}_2 = \frac{y_1 - y_2}{3}}$$

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These are the LSEs

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

A-2 a) ~~Var(X)~~ $\text{Var}(X) = \frac{(b-a)^2}{12}$ $E[X] = \frac{b+a}{2}$ $E[X^2] = \frac{b^2+a^2+ab}{3}$

When $X \sim U(a, b)$

~~U₁ & U₂ are independent~~

So, $\text{Cov}(U_1, U_2) = 0$ ~~$\text{Cov}(U_1, U_2) = \text{Cov}(U_1, U_1 + U_2)$~~

$\text{Var}(U_1) = \text{Var}(U_2) = \frac{1}{12}$ $E(U_1) = E(U_2) = \frac{1}{2}$

~~Corr~~

$\rho_{11} = \rho_{22} = \rho_{33} = 1$

as $\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var } X} \sqrt{\text{Var } Y}}$

For $X = Y$

$\rho = \frac{\text{Cov}(X, X)}{\text{Var}(X)} = 1$ ($\text{Cov}(X, X) = \text{Var}(X)$)

$\rho_{12} = \frac{\text{Cov}(U_1, U_2)}{\sqrt{\text{Var}(U_1)} \sqrt{\text{Var}(U_2)}} = 0$

$\rho_{13} = \frac{\text{Cov}(U_1, U_1 + U_2)}{\sqrt{\text{Var}(U_1)} \sqrt{\text{Var}(U_1 + U_2)}} = \frac{E(U_1^2 + U_1 U_2) - E(U_1)E(U_1 + U_2)}{\sqrt{\frac{1}{12}} \times \sqrt{\frac{1}{12} + \frac{1}{12}}}$
 $\hookrightarrow$ As $\text{Cov}(U_1, U_2) = 0$
 $= \frac{\frac{1}{3} + \frac{1}{2} \times \frac{1}{2} - \frac{1}{2}(\frac{1}{2} + \frac{1}{2})}{\frac{\sqrt{2}}{12}}$

$$= \left(\frac{1}{3} + \frac{1}{4} - \frac{1}{2} \right) \frac{12}{\sqrt{2}}$$

$$= \left(\frac{7}{12} - \frac{6}{12} \right) \frac{12}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$\rho_{23} = \rho_{13}$ as just u_1, u_2 terms are interchanged

So, $\bar{P} = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} \rightarrow \text{Correlation Matrix}$

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b) Now we need eigen value of $\bar{P}$

$$\text{So, } \det \begin{bmatrix} 1-\lambda & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{bmatrix} = (1-\lambda) \left[(1-\lambda)(1-\lambda) - \frac{1}{2} \right] + \frac{1}{\sqrt{2}} \left[-(1-\lambda) \frac{1}{\sqrt{2}} \right] = 0$$

$$\Rightarrow (1-\lambda)^3 - \frac{1}{2}(1-\lambda) - \frac{1}{2}(1-\lambda) = 0$$

$$\Rightarrow (1-\lambda)^3 - (1-\lambda) = 0$$

$$\Rightarrow (1-\lambda) [(1-\lambda)^2 - 1] = 0$$

$$\Rightarrow (1-\lambda)(2-\lambda)(-\lambda) = 0$$

$$\lambda = 0, 1, 2$$

For 1st principal component we take largest eigenvalue which is 2

$$\text{So, } \lambda_1 = 2$$

Continued at Pg 9

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

A-3 a) i) Least distance is between 1) and 2)

So, cluster will be formed of (1) & (2)

	(1 2)	3	4
(1 2)	0		
3	11	0	
4	5	3.5	0

$11 = \max(a_{13}, a_{23})$
 $5 = \min(a_{14}, a_{24})$

Complete Linkage $\rightarrow$ max distance from cluster is used

Now least distance is of 3) and 4)

So, a new cluster of (3, 4) is made

	(1 2)	(3 4)
(1 2)	0	
(3 4)	11	0

$11 = \max(a_{(12)3}, a_{(12)4})$

Now last cluster is (1, 2, 3, 4) which has all the elements together

ii) Again first cluster is between (1) and (2)

	(1 2)	3	4
(1 2)	0		
3	6.5	0	
4	4	3.5	0

$$a_{(12)3} = \frac{11+2}{2}$$

$$= 6.5$$

$$a_{(12)4} = \frac{5+3}{2}$$

$$= 4$$

~~Now~~ Average Linkage $\rightarrow$ Avg of distance from cluster

Now least distance is between (3) and (4)

So, new cluster b/w (3) & (4)

So,

	(1 2)	(3 4)
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(1 2)	(3 4)
0	
	0
(3 4)	5.25

$$a_{(12)(34)} = \frac{6.5+4}{2}$$

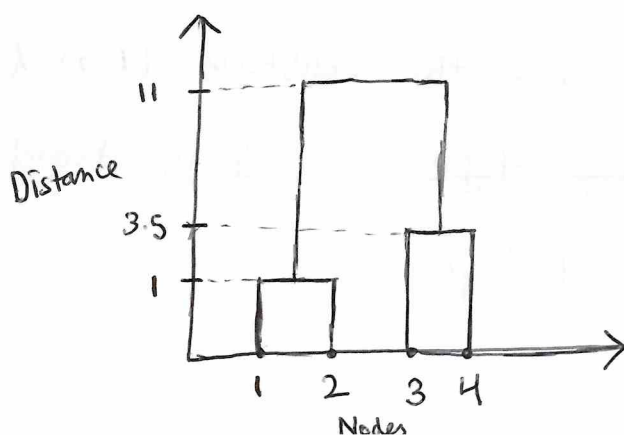
$$= \frac{10.5}{2}$$

$$= 5.25$$

Ob

So, last cluster is (1, 2, 3, 4)

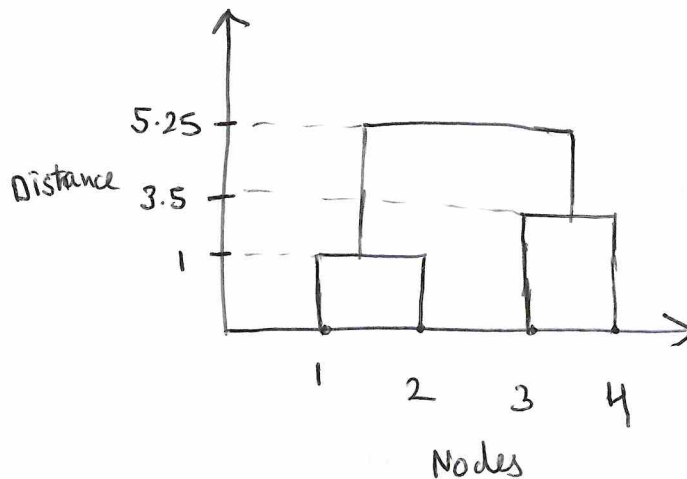
b) Dendrogram for complete linkage



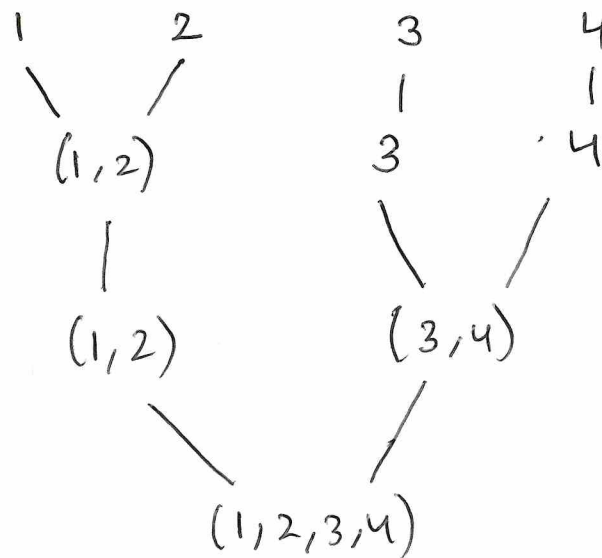
Continued at Pg 8

A-3 b) contd.

Dendrogram for Avg Linkage



So, the formation of cluster in both type of ~~cluster~~ linkage is similar and it is:



But the last distance b/w the clusters (1,2) & (3,4) is different in both linkage type i.e., 11 in complete linkage and 5.25 in Average linkage

A-2 contd

So, we need eigen vector e_1

$$\begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2a \\ 2b \\ 2c \end{bmatrix}$$

$$\Rightarrow a + \frac{c}{\sqrt{2}} = 2a$$

$$b + \frac{c}{\sqrt{2}} = 2b$$

$$\frac{a}{\sqrt{2}} + \frac{b}{\sqrt{2}} + c = 2c$$

$$\Rightarrow c = \frac{a}{\sqrt{2}} + \frac{b}{\sqrt{2}} ; b = \frac{c}{\sqrt{2}} ; a = \frac{c}{\sqrt{2}}$$

$$\text{So, } a:b:c = \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{2}} : 1$$

$$\frac{\sqrt{1 + \frac{1}{2} + \frac{1}{2}}}{\sqrt{2}}$$

$$\text{So, } e_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 1 \end{bmatrix} \rightarrow \text{Normalised}$$

$$\text{So, } PC_1 = \frac{x_1}{\sqrt{2}} + \frac{x_2}{\sqrt{2}} + \frac{x_3}{\sqrt{2}}$$

We need $\text{Var}(PC_1)$

~~$$\text{Now } x_3 = x_1 + x_2$$~~

~~$$\begin{aligned} \text{So, } \text{Var}(PC_1) &= \text{Var}\left(\frac{x_1}{\sqrt{2}} + \frac{x_2}{\sqrt{2}} + \frac{x_3}{\sqrt{2}}\right) = \frac{1}{2} \text{Var}(x_1 + x_2 + x_3) \\ &= \frac{1}{2} (1 + 1 + 1) = \frac{3}{2} = 1.5 \end{aligned}$$~~

~~$$\text{Var}(PC_1) = \frac{1}{4} \left(\frac{1}{12} + \frac{1}{12} \right) + \frac{1}{4} \left(\frac{1}{12} \right) + \frac{1}{2} \left(\frac{2}{12} \right) + 2 \left(\frac{1}{12} \right) + 2 \left(\frac{1}{12} \right) + 2 \left(\frac{1}{12} \right)$$

$$\frac{1}{2} \left(\frac{1}{12} \right) + \frac{4}{12} + \frac{3}{12} + \frac{1}{2}$$~~

$$\begin{aligned} \text{Var}(PC_1) &= \text{Var} \left(\frac{X_1}{2} + \frac{X_2}{2} + \frac{X_3}{\sqrt{2}} \right) \\ &= \text{Var} \left(\frac{X_1}{2} + \frac{X_2}{2} \right) + \text{Var} \left(\frac{X_3}{\sqrt{2}} \right) + 2 \text{Cov} \left(\frac{X_1 + X_2}{2}, \frac{X_3}{\sqrt{2}} \right) \\ &= \frac{1}{4} \times \left(\frac{1}{12} + \frac{1}{12} \right) + \frac{1}{2} \left(\frac{1}{12} \right) + 2 \text{Cov} \left(\frac{X_1 + X_2}{2}, \frac{X_3}{\sqrt{2}} \right) \end{aligned}$$

$$\mathbb{E} \left[\frac{X_1 X_3 + X_2 X_3}{2\sqrt{2}} \right] - \mathbb{E} \left[\frac{X_1 + X_2}{2} \right] \mathbb{E} \left[\frac{X_3}{\sqrt{2}} \right]$$

$$\frac{1}{2\sqrt{2}} \left[\frac{7}{12} + \frac{7}{12} \right] - \frac{1}{2\sqrt{2}} \left[\frac{2}{12} \times \frac{1}{12} \right]$$

$$= \frac{1}{2\sqrt{2}} \left[\frac{7}{6} - \frac{2}{144} \right] = \frac{1}{2\sqrt{2}} \left[\frac{7}{6} - \frac{1}{72} \right]$$

 $u_1^2 + u_1 u_2$

$$= \frac{83}{144\sqrt{2}} = \boxed{0.40757}$$

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So, $\boxed{\text{Var}(PC_1) = 0.40757}$